

FULL SMS USER MANUAL

Table of Contents

Table of Contents	1
1 Introduction	2
1.1 Who Uses Full SMS	2
1.2 Key Features of Full SMS	2
2 Technical Specifications	2
3 System Requirements	2
4 Get Started	3
4.1 Log In to the System	3
4.2 Create a Workspace	3
4.3 Upload Data	4
5 Analyze Data	5
5.1 Use the Intensity Tab	5
5.2 Use the Grouping Tab	5
5.3 Use the Lifetime Tab	6
5.4 Use the Raster and Spectra Tabs	7
6 Manage Plugins	7
7 Manage the Account	8
8 Recommendations	8
9 Frequently Asked Questions	8
10 Change Log	8

1 Introduction

Full SMS is a web-based software platform for Single-Molecule Spectroscopy. You use it to analyze HDF5 measurement files from a web browser. It replaces the older desktop software to make access easy.

1.1 Who Uses Full SMS

Researchers from the University of Pretoria Biophysics Group use this system to analyze data. Collaborators also use it to look at shared datasets.

1.2 Key Features of Full SMS

- You access the software from a web browser without a local installation.
- The system uses secure authentication and isolated workspaces.
- You can upload HDF5 files and convert .pt3 or .csv files.
- The system does change point detection and hierarchical clustering.
- The system does fluorescence lifetime fitting and correlation analysis.
- You can add custom analysis plugins.

2 Technical Specifications

- **Frontend:** The user interface uses Next.js 14 and Tailwind CSS.
- **Backend:** The server uses FastAPI and Python 3.12.
- **Database:** The system uses PostgreSQL and Supabase.
- **Task Queue:** The system uses Celery and Redis for background tasks.
- **Plugin Execution:** Plugins run in an isolated Oracle Cloud environment.
- **Typography:** The interface uses Public Sans and JetBrains Mono fonts.

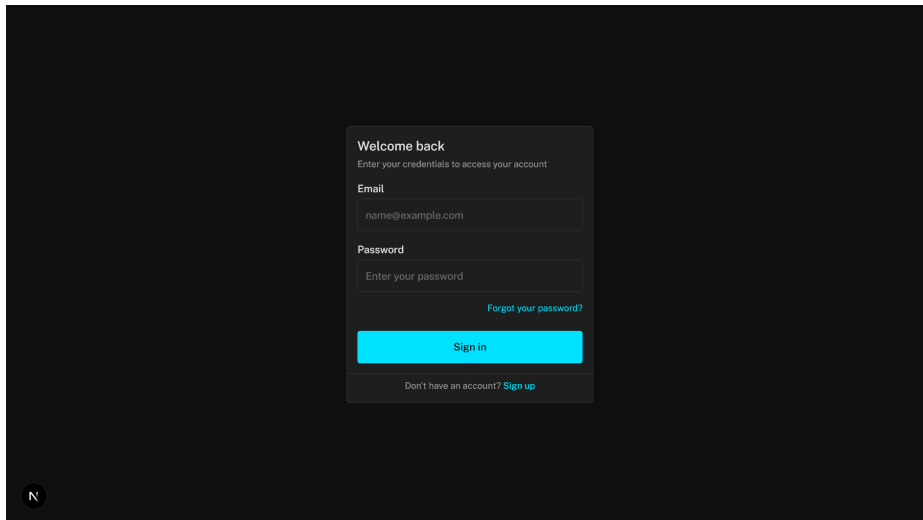
3 System Requirements

- You must have a modern web browser.
- You must have a stable internet connection.
- You must use .pt3, .csv, .h5, or .hdf5 data files.

4 Get Started

4.1 Log In to the System

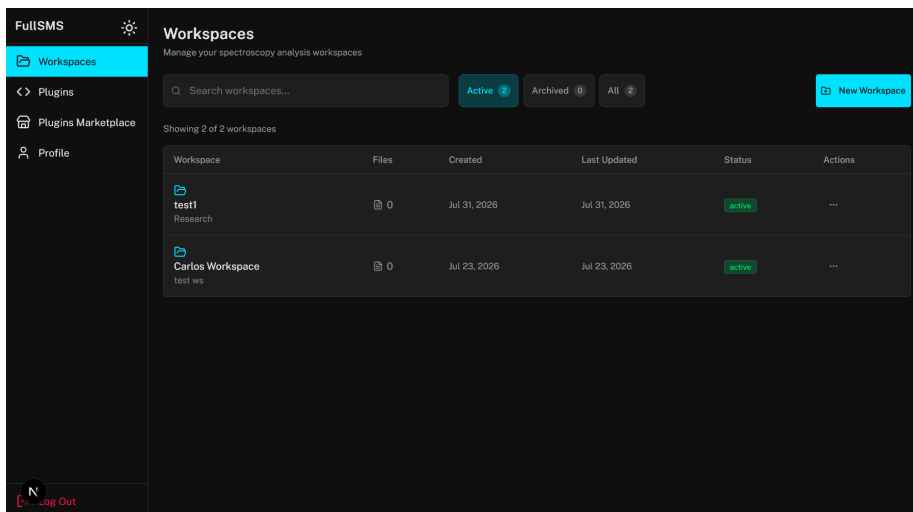
1. Open the Full SMS web address in your browser.



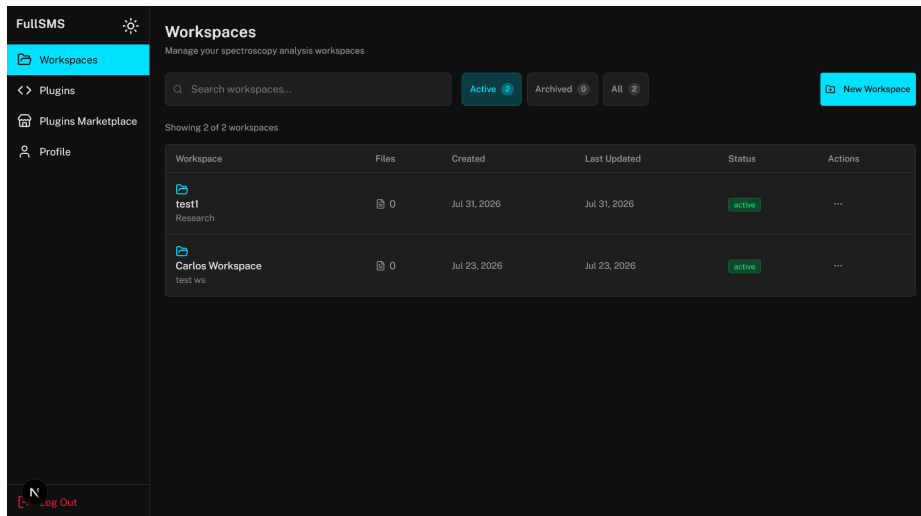
2. Type your registered email address and password.
3. Click the Login button to access your dashboard.

4.2 Create a Workspace

1. Click the New Workspace button on the dashboard.



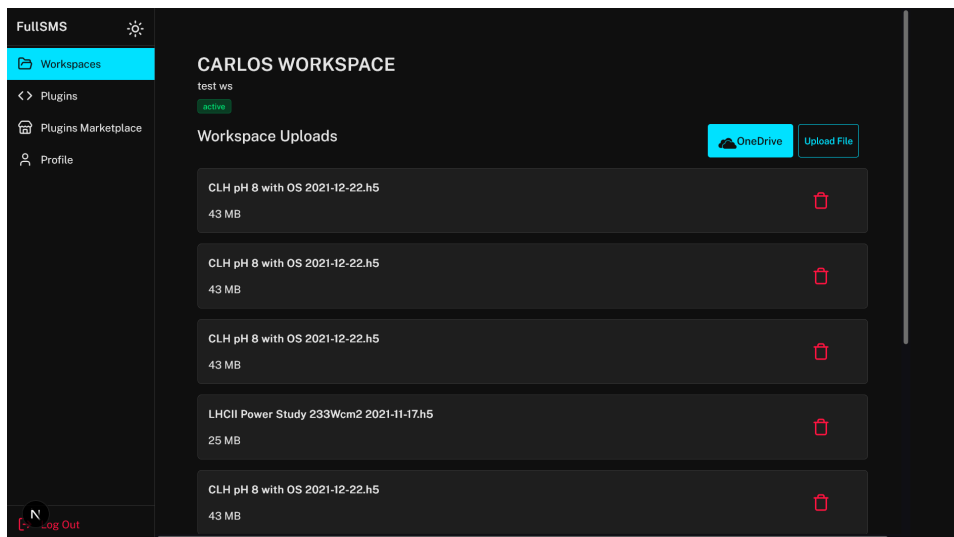
2. Type a name and a description for the workspace.



3. The system saves the active workspace to your dashboard.

4.3 Upload Data

1. Open a workspace.
2. Click the Upload File button.

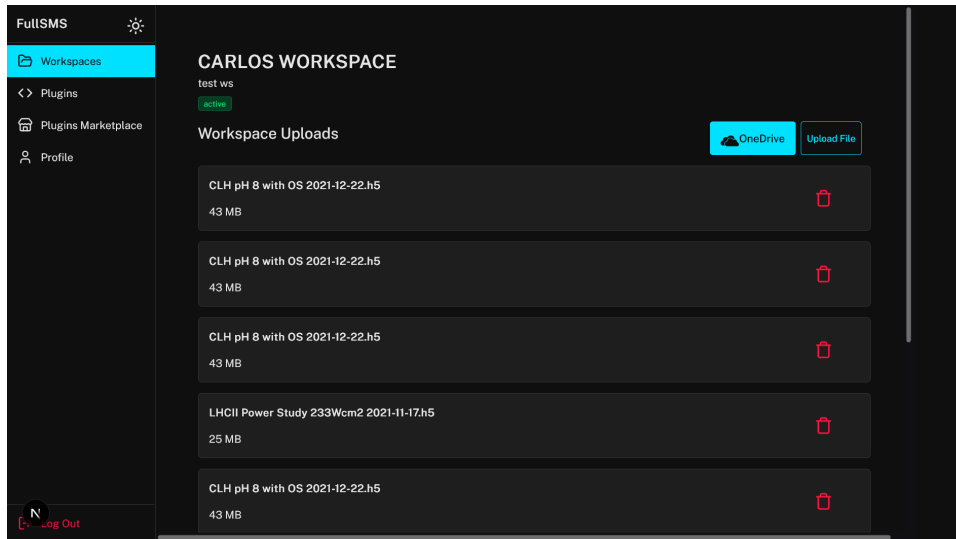


3. Select a measurement file from your computer.
4. Wait for the system to read the file.

5 Analyze Data

5.1 Use the Intensity Tab

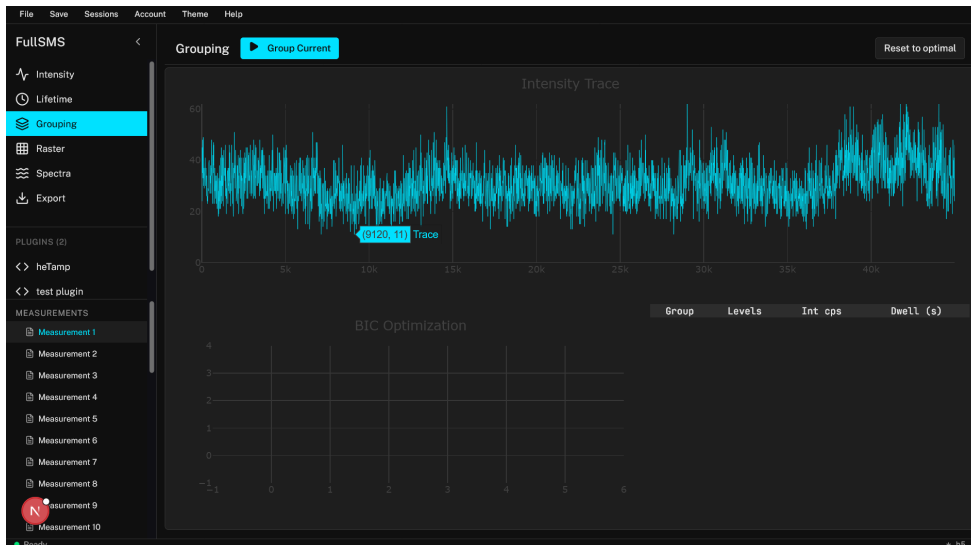
1. Click the Intensity tab.



2. Look at the zoomable time-trace plot.
3. Type the binning width parameters.
4. Start the Change-Point Analysis (CPA).
5. Wait for the background task to complete.

5.2 Use the Grouping Tab

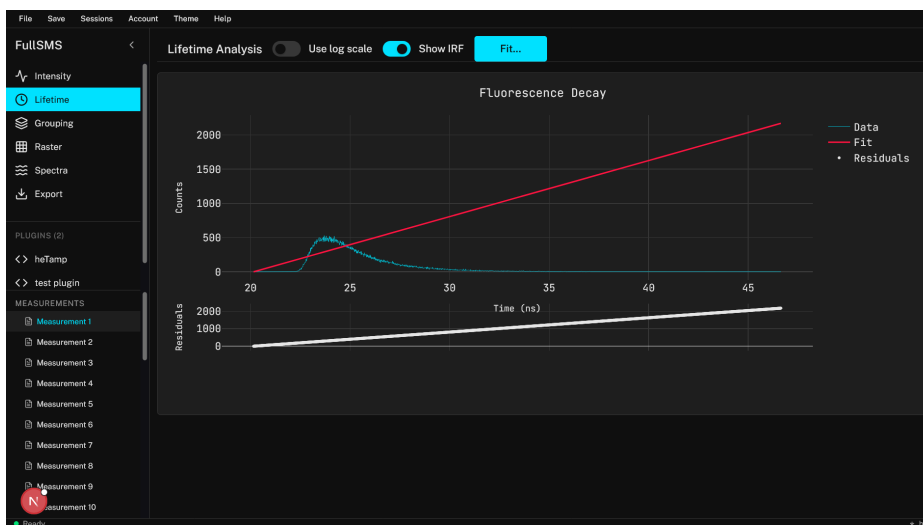
1. Click the Grouping tab.



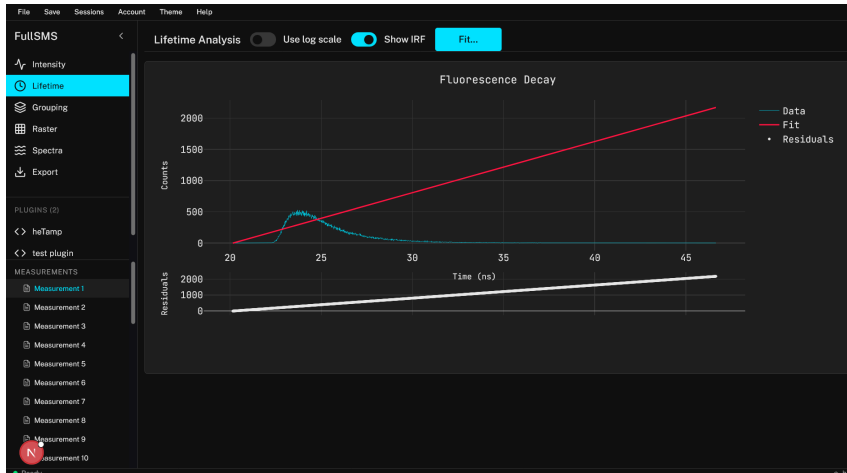
2. Start the clustering analysis.
3. Look at the Bayesian Information Criterion (BIC) plots.

5.3 Use the Lifetime Tab

1. Click the Lifetime tab.



2. Start the multi-exponential decay fit.



3. Look at the fitted decay curves.

5.4 Use the Raster and Spectra Tabs

1. Click the Raster tab to look at 2D raster scans.
2. Click the Spectra tab to look at spectra data.

6 Manage Plugins

1. Click the Plugins tab to see the your plugins

Plugin	Version	Parameters	Outputs	Status	Marketplace	Updated	Actions
heTamp heatmap test	v1.0.0	2	1	Enabled	Not Submitted	Jul 31, 2026	...
test plugin	v1.0.0	1	1	Enabled	Not Submitted	Jul 31, 2026	...

2. Click the New Plugin button to create a new plugin.
3. Follow the on screen guides to help you write your first plugin.

7 Manage the Account

1. Open the profile settings.
2. Update your username or password.
3. Save your default analysis parameters.

8 Recommendations

- Save your analysis session before you close the browser.
- Use the batch export function to save time.

9 Frequently Asked Questions

- **Can I share my data?** Yes. By sharing access to files and folders on your cloud storage providers.
- **Do I need to install Python?** No. The software operates entirely in the web browser.

10 Change Log

Version	Date	Changes
1.0	2026-07-31	Initial document release.
1.1	2025-09-03	Demo 3

